Rethinking case attraction on Ancient Greek infinitive clauses

Caio B. A. Geraldes*
<caio.geraldes@usp.br>

June 25, 2025

1 Introduction

Good morning. Nominal predicates and adjuncts modifying subjects of complement infinitive clauses in Ancient Greek (AG) usually agree in case with their domain local subject, which has morphological accusative case assigned in the *Accusativus cum Infinitivo* (*AcI*) construction. When the subject of the infinitive is not overtly expressed but the main clause has a dative or genitive object which shares the reference of the infinitive subject, these predicates or adjuncts may either occur in the accusative or "be attracted" to the case of the main clause's object.

(1) a. συμβουλεύει **τῷ Ξενοφῶντι ἐλθόντα** εἰς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advices Xenophon to go to Delphi and ask the god about the travel. (Xen. *Anab.* 3.1.5)

b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τἀληθῆ.
He allowed me to go and speak the truth in front of you. (Xen. *Hell.* 6.1.13)

The literature on the subject, on the more philological side, from Port-Royal (Lancelot 1658, 1709) up to the Cambridge Grammar (Boas et al. 2019), have focused on listing the contexts in which attraction may occur, whereas the transformational and minimalist approaches of Lakoff (1970), Andrews (1971), Quicoli (1982), Spyropoulos (2005), Tantalou (2003), Sevdali (2007, 2013) have

*Fundação de Amparo à Pesquisa do Estado de São Paulo.

mainly occupied themselves with proposing the structure that would allow the phenomenon to occur. What factors influence the decision between either resolution in (1) remains, however, largely unknown.

[Self-deprecatingly] I myself have, in the last edition of this same event, tried to present the distributional patterns that make the attraction, (1-b), more likely, showing that there is strong correlation between attraction and the classes of the matrix and infinitive verbs and the relative distance and position of the controller and target of agreement.

Furthermore, I argued that these factors were *proxies* to unobserved values, namely semantic and pragmatic features of the discourse setting, and that even in the absence of seen contrasting factors, precisely the case in the pair above (1), the unseen discourse factors would be still causing one resolution to be at the very least more likely.

[Introductingly] I wasn't, however, quite convinced about neither the methods, the data, and, consequently, the quality of the results at which I had arrived. The main reason for this distrust is the fact that there could be interaction between the factors analysed for which the methods employed were not well designed to account, at least not with the amount of data available. A secondary motive was the classic adage "correlation is not causation".

To solve these shortcomings, I have collected more data from a larger *corpus* but more importantly swapped the quantitative framework for one capable of dealing with multivariate interactions, namely from "out of the shelf" statistical tests to generalized linear models, and, I believe more importantly, to for one capable of expressing the causality behind the statistical correlations.

For this to work, however, we must take some steps back and build a causal model for what is the linguistic nature of case attraction and how properties of the sentence and its constituents might favor its occurrence. The following is a walk-through the construction of this causal model and its results.

2 The data

Before stepping into the framework and analysis proper, a brief word on the empirical data supporting this research and any conclusion coming from it. The dataset is composed of 343 sentences drawn from the Attic literary *corpus* and Herodotus. The sentences were extracted from the Diorisis Corpus (Vatri and McGillivray 2018, 2020) with help of the project's querying tool, Diorisis Search (Vatri 2020-2022). Although comprehensive, the selection is by no means exhaustive.

Each sentence has been annotated for values of interest, including authorship, literary genre, dialect as well as class of matrix and infinitive verb, class of controller and target, distance between constituents, etc. Parts of the annotation processes were conducted programmatically and later checked manually.

Every model used in this work was processed using R (R Core Team 2013), the packages tidyverse (Wickham et al. 2019), dagitty (Textor et al. 2016), ggdag (Barrett 2025), and rethinking (McElreath 2020), as well as the software Stan (Stan Development Team 2025a,b).

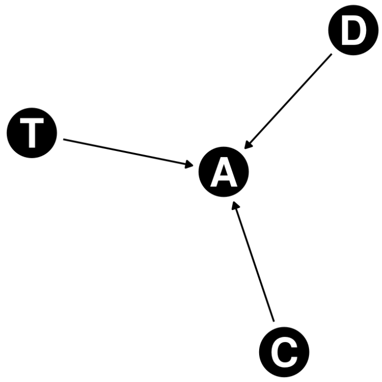
3 Nature of attraction

In the most abstract way possible, case attraction in infinitive clauses, exemplified by (1-b), is when two constituents, a controller and a target, produced in different domains of a sentence, one of the domains encompassing the other, covary in an specific feature, namely $[Case]$. This satisfies the definition of agreement in Corbett's typological assessment of the phenomenon across languages (2006, pp. 4–7).

The lack of covariation of the $[Case]$ feature, seen in (1-a), is generally understood as result of the application of **default** case concord rules of Ancient Greek. This assumes that the subject of infinitive clauses in Greek is always marked as accusative, even when not overtly expressed, and that convert constituents are also marked for $[Case]$ and its case is also available for nominal concord.

The facts that the target has a choice of controller (either the overt oblique or de convert accusative) for agreement, that $[Case]$ is a non-lexical features, that the agreement is optional, and that the domain is non-local favours the interpretation that the agreement type of case attraction is less canonical than the general agreement type of Ancient Greek, following the criteria for canonical agreement in the same work of Corbett (2006, pp. 8ff.).

If this assessment is sound, it is expected that attraction and non-attraction behave as competing agreement resolutions, the former being less canonical and thus more likely to be conditioned. Conditions concerning the semantic and pragmatic properties of the controller, its precedence in relation to the target and the class of predication (i.e.: class of the domain) have been identified by Corbett. Although properties of the target were not shown to be favouring any resolution in agreement mismatches in Corbett's survey, case attraction might be one example of it, and I will hypothesize that properties of targets are competing causes of non-canonical resolutions. A model of agreement in this framework may be represented as a direct acyclic graph, in which **A**ttraction ($\simeq$ Agreement) is caused by features of the **C**ontroller, **D**omain and **T**arget:



These Controller, Domain and Target nodes represent clusters of linguistic factors, some of which we may observe directly (part of speech of the target), some might be implied (animacy) and some may remain unobserved (discourse function, as Ancient Greek does not overtly marks focuses or topics in the morphosyntax, at least not dependably). When an observed factor is included in the model, we split the cluster into two nodes: the one of observed factor and that containing rest of unobserved features.

The three factors previously shown to correlate to case attraction represent each of these nodes:

- Copular infinitives imply a different domain of agreement, as they select different predication classes;
- Main verbs with semantic modality imply a different type of controller, as they have higher subjecthood;
- Word order and distance imply different status for the target, as targets more to the left of the sentence are both structurally closer to the controller and have a higher communicative function.

4 The domain of attraction

Firstly, Kühner and Gerth (1898) and others have associated case attraction with sentences in which the infinitival clause is a copular infinitive, einai or genesthai. Should the class of the infinitive favour either resolution of agreement? Probably not directly, as there is no reason linguistically it would. Copular infinitives, however, have a double effect on the agreement "causes":

1. they filter the possible relations between controller and target, as those are now more likely in primary predication, rather than secondary or appositive relation;
2. this then delimits the option of target, making it more likely that they are nouns or adjectives, rather than participles.

The these are known to affect agreement resolutions, and a couple of Hierarchies have been proposed to predict when the less canonical resolution (also called semantic agreement) is more likely to take place. The Agreement Hierarchy (Corbett 1979) and Predicate Hierarchy (Comrie 1975) have also been combined into the Agreement and Predicate Hierarchy shown in (1):

(1) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):

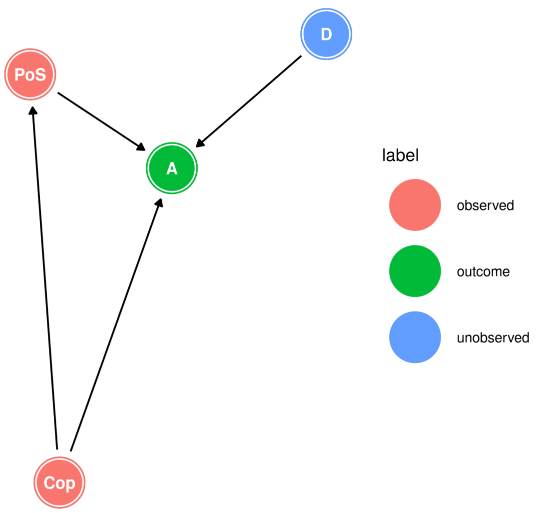
Noun
↗
Adjective
↗
Participle
↗
Attributive → Predicative Verb → Relative Pronoun → Personal Pronoun

- ⟵⟶ +

Likelihood of agreement with semantic justification

Adding this to our causal model, we may say that **A**ttraction is caused both by whether or not the relation between the controller and target is copular, and by the Part of Speech of the target, which is itself "caused" by the type of relation (copulas select more nominal predicates on average), and other unobserved properties of the **D**omain may also cause Attraction.

By applying the do-calculus (Pearl 2009), we learn that: the direct effect of the part of speech class requires us to adjust the statistic model by C, so that we block the non-causal path $P \leftarrow C \rightarrow A$.



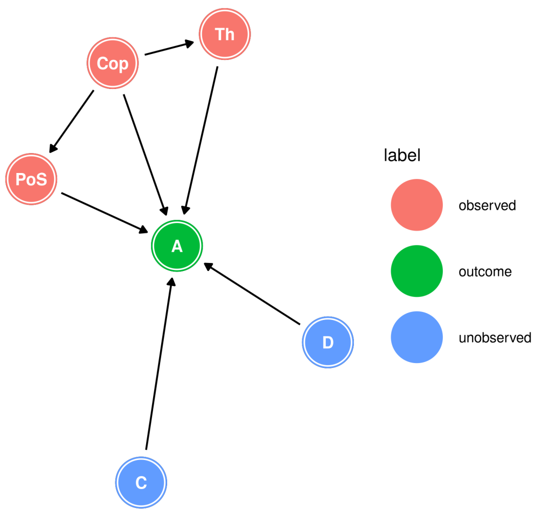
5 The controller of attraction

Another association is that between case attraction and impersonal verbs. I have, however, shown that this association does not exist with every single impersonal verb, and that the verb δοκέω almost never heads a sentence with case attraction. The impersonal verbs which are actually associated with case attraction are those carrying modal semantics, such as ἐξαρκεῖ and ἔξεστί. Is it likely that the nature of the matrix verb directly affects the choice of agreement? I don't think so, but it may indirectly do, by defining which features the controller of agreement, the oblique object, has. These verbs have very little semantic information and the theta role and grammatical function of their oblique object seems to depend on the infinitive verb. We may say that σοι in (1), for example, is the subject of the sentence (a quirk subject of sorts) and has the thematic role of agent, much in contrast if the sentence was constructed with a verb of order or advice like συμβουλέυω, to which the oblique represents a recipient of the action, and if the sentence is constructed by δοκεῖ, to which the oblique represents an experiencer, "the one who believes X to be better to do".

(1) ἀλλ' ἐξαρκῆ σοι τὰ ἔργα αὐτῶν ὁρῶντι σέβεσθαι καὶ τιμᾶν τοὺς θεούς.
But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

This, however, only holds if the infinitive verb is not copula, otherwise, the thematic role is also that of an experiencer, meaning that **Th**ematic Role is a sister node of **P**art **o**f **S**peech, as it is "caused' by **C**opula.

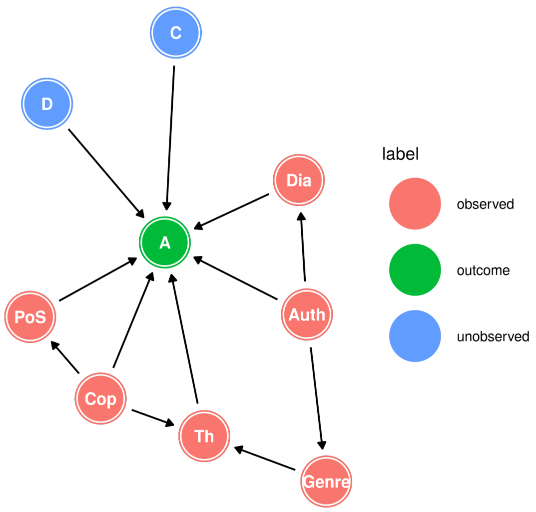
Adding to our previous DAG, the model will look like:



6 Intermezzo: Authors and dialects

I have argued previously that, in the corpus of Herodotus, Plato and Xenophon, the verbs of modal semantics were preferred by the authors when writing in the philosophical dialogue genre, and that this had implications if we were to estimate the effect of authorship or dialect on case attraction. If modality verbs affect the likelihood of attraction by means of selecting the thematic role of the controller, the author's idiosyncrasies in the use of attraction or their dialect base likelihood of using attraction would be biased in the data by the effects of the selection of the main verb in sentences where attraction may appear.

We must thus add **Auth**orship, **Genre** and **Dia**lect to our causal model.



These last couple of additions have the following consequences:

- to estimate the direct effect of **Th**ematic role, we must control for **Copula** and either by **Genre** or **Auth**orship;
- to estimate the direct effect of **Cop**ula, we must control for **P**art **of** **S**peech, **Th**ematic role, and either by **Genre** or **Auth**orship;
- to estimate the direct effect of **Dia**lect, we must control for **Auth**orship;
- to estimate the direct effect of **Auth**orship, we must control for **Dia**lect and by either **Genre** or **Cop**ula and **Th**ematic role.

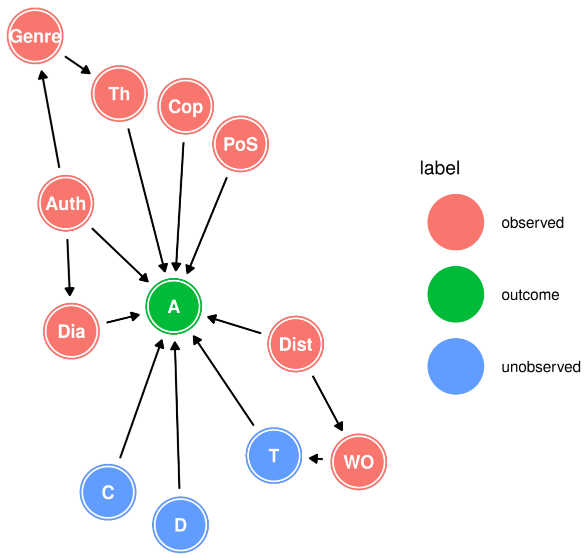
7 The target of attraction

Finally, it has been long been proposed that case agreement is more likely when controller and target are very close to each other, and the likelihood diminishes as it grows. Furthermore, precedence of the target in relation to the controller seems to demand case attraction, such as in (1).

(1) ὡς **ἀγαθοῖς** τε **ὑμῖν** προσήκει εἶναι.
That you may be good people. (Xen. *Anab.* 3.2.11)

Word order is highly associated in Ancient Greek with an important condition for non-canonical agreement: communicative function. [1]. Although associated with, however, word order does not completely determines communicative function of the target: it is not enough to guarantee that a certain constituent is a topic or focus. In other words, Word Order would only affect case attraction indirectly, by 'causing' the communicative function of the target, function which is, however, unobservable. This means that the effect of word order over case attraction is statistically impossible to infer without a way to measure the communicative function of the target.

The effect of distance, on the other hand, is measurable, so long as we control for its effect on word order: distance changes word order, so we must block the non-causal path that flows from distance to attraction going through word order and communicative function. In the DAG:



[1] Allan (2012), Dik (1995, 2007), and Matić (2003)

8 Analysis

Fortunately, we don't need many different models to estimate the effects of many of the factors of interest. By applying the do-calculus, so long as we control for Word Order, we can estimate the direct effects of: 1. Part of Speech of the target; 2. Copular infinitive (i.e.: class of predication); 3. Thematic role of the oblique object; 4. Dialect; 5. Authorship; and 6. Distance. The following generalized linear model allows us to do so:

$$
\begin{aligned}
\text{A} &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha + \beta_{\text{Cop}}\text{Cop}_i + \beta_{\text{Dist}}\text{Dist}_i + \beta_{\text{WO}}\text{Prec}_i \\
&\quad + \beta_{\text{PoS}} \sum_{j=0}^{PoS_i - 1} \delta_{j_{\text{PoS}}} + \beta_{\text{Th}} \sum_{j=0}^{Th_i - 1} \delta_{j_{\text{Th}}} \\
&\quad + \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i} \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta_{\text{Cop}}, \beta_{\text{Dist}}, \beta_{\text{Prec}} &\sim \text{Normal}(0, 1) \\
\beta_{\text{PoS}}, \beta_{\text{Th}} &\sim \text{LogNormal}(0, 1) \\
\delta_{\text{PoS}}, \delta_{\text{Th}} &\sim \text{Dirichlet}(2) \\
\beta_{\text{Auth}} &= z_{\text{Auth}} * \sigma_{\text{Auth}} \\
\beta_{\text{Dia}} &= \overline{\beta}_{\text{Dia}} + z_{\text{Dia}} * \sigma_{\text{Dia}} \\
\overline{\beta}_{\text{Dia}}, z_{\text{Auth}}, z_{\text{Dia}} &\sim \text{Normal}(0, 1) \\
\sigma_{\text{Auth}}, \sigma_{\text{Dia}} &\sim \text{Exponential}(1)
\end{aligned}
$$

A brief note on priors: Those are fairly flat, as there is no reason to believe we have weak or strong relationships in agreement phenomena. The effects of Part of Speech and Thematic Role (the scary ones with sums) are modeled as Ordered Categorical Predictors using the Dirichlet distribution. This is to represent the generally agreed idea that typological hierarchies have "cutoff" points in between the nodes and that their effect increases "monotonically" as we traverse between the nodes. The first node of each hierarchy will have the base effect β_X, and the following nodes will "accumulate" effect. Using Part of speech as an example, β_{PoS} is the general effect of the part of speech over the likelihood of Attraction, and δ is a vector with values denoting how much going up from Participle to Adjective to Noun increases β_{PoS}. Participles do not change the effect, Adjectives add δ_1 and Nouns adds both the effects $\delta_1 + \delta_2$.

The effects Authorship and Dialect have non-centered priors to preform partial pooling, so that we are able to estimate how much variation there is between each of their values.

8.1 Copula and Distance

The most generally associated factors with attraction, namely the infinitive being a copula and the distance between controller and target show the expected effects:

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	2.03	0.39	1.42	2.66	1
β_{Dist}	-0.22	0.04	-0.29	-0.16	1

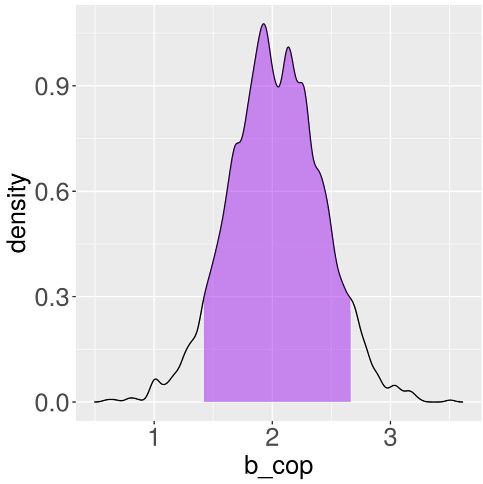


Figure 1: Posterior distribution of β_{Cop} with the 89% Highest Posterior Density Interval.

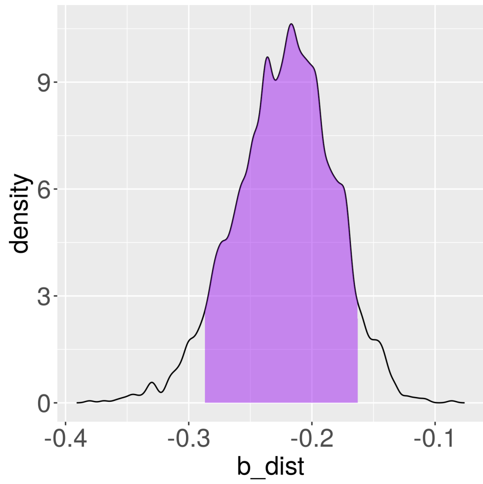


Figure 2: Posterior distribution of β_{Dist} with the 89% Highest Posterior Density Interval.

The estimate effect for Distance is quite precise, and, to offer an intuition of what these values mean, the model predicts that by the time the controller and target are 8 words apart, on average, the fact that the infinitive is a copula is no longer relevant, i.e., the attraction is no longer a more likely outcome. If our causal model is sound, this is an important result for the typology of agreement in general, as the evidence from other languages predict that higher distances between controller and target would favor non-canonical agreement resolutions (e.g.: modifiers of committee nouns in English are more likely to stand in the plural when far away from their controller, as shown by Levin (2001, p. 98)).[2].

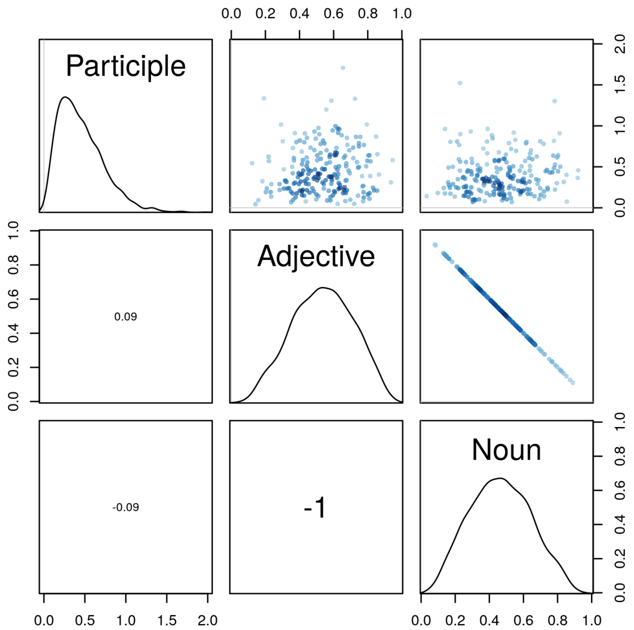
[2] Other examples in Corbett (2006, p. 235)

8.2 Part of Speech and the Predicate Hierarchy

The estimated effects are described in the following table and plot.

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	0.45	0.28	0.11	0.95	1
δ_{Adj}	0.53	0.18	0.23	0.82	1
δ_{Noun}	0.47	0.18	0.18	0.77	1

PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	$\beta_{\text{PoS}} * 0$	0.00	0.00	0.00	0.00	NA
Adjective	$\beta_{\text{PoS}} * \delta_{\text{Adj}}$	0.25	0.18	0.05	0.60	1
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$	0.45	0.28	0.11	0.95	1



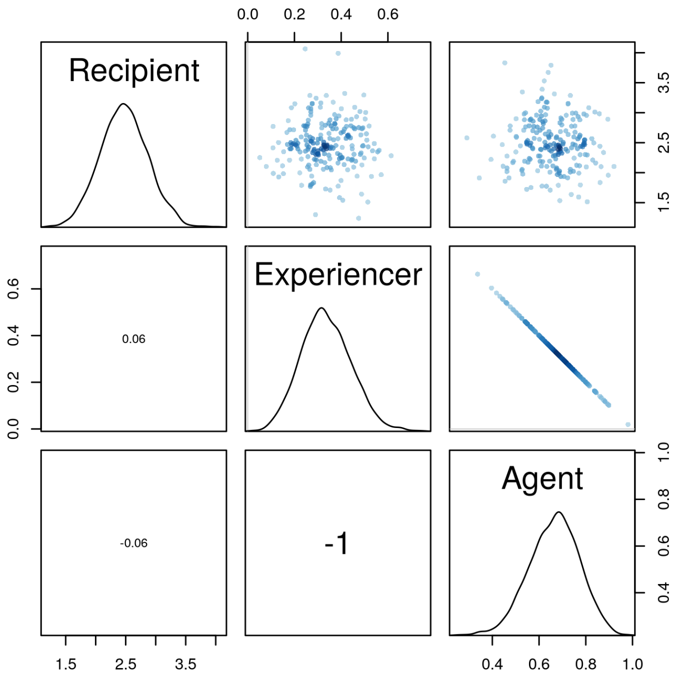
To put it into words, the data favours the idea that the Agreement and Predication Hierarchy has a reasonable effect over case attraction: as the target becomes more "nominal", the likelihood of attraction increases seemingly monotonically. There is still a fair amount of uncertainty on how strong the effect is, however.

8.3 Thematic Role of the Controller

The estimated effects are described in the following table and plot.

Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	2.47	0.40	1.84	3.12	1
δ_{Exp}	0.34	0.11	0.17	0.51	1
δ_{Agent}	0.66	0.11	0.49	0.83	1

Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	$\beta_{\text{Th}} * 0$	0.00	0.0	0.00	0.00	NA
Experiencer	$\beta_{\text{Th}} * \delta_{\text{Exp}}$	0.83	0.3	0.38	1.34	1
Agent	$\beta_{\text{Th}} * (\delta_{\text{Exp}} + \delta_{\text{Agent}})$	2.47	0.4	1.84	3.12	1



Again, the results confirm the hypothesis that attraction is more likely to occur when the oblique object of the matrix clause is more "canonically" a subject. The effect is quite strong and the model is quite certain that when the matrix verb encodes modality and the thematic role of 'Agent' is assigned by the infinitive verb, there is a big increase in the likelihood of attraction.

8.4 Dialect and Authorship

The model estimates that Attic and Jonic are quite different on how often they employ case attraction, Jonic being predicted to use it much less often when all the other factors are the same. Although the model can not estimate a precise value (the standard deviation is on the higher end), it is very confident of the contrast. It also predicts that dialects should have a great amount of variation, as denoted by the parameter σ_{Dia}.

Effect	mean	sd	5.5%	94.5%	rhat
$\overline{\beta}_{\text{Dia}}$	-0.49	0.84	-1.78	0.90	1.00
σ_{Dia}	1.20	0.78	0.30	2.67	1.01
β_{Attic}	-0.16	0.89	-1.55	1.28	1.00
β_{Jonic}	-1.59	1.01	-3.19	-0.02	1.00
$\beta_{\text{Attic}} - \beta_{\text{Jonic}}$	1.43	0.69	0.26	2.49	NA

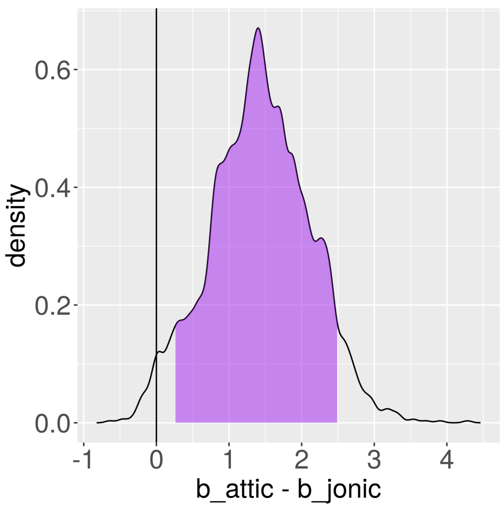
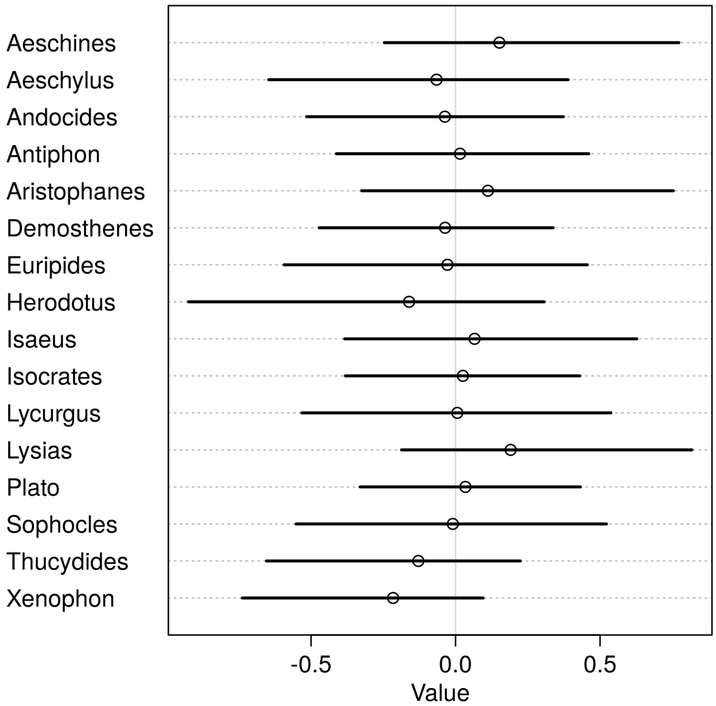


Figure 3: Posterior distribution of the difference between β_{Attic} and β_{Jonic} with the 89% Highest Posterior Density Interval.

Finally, a word on authorship. After controlling for dialect and preferred choice of matrix verb, most of the authors are predicted to similar in their use of case attraction, with a curious tendency of Xenophon to be, on average, less likely to employ it.



References

Allan, R. (2012). "Clause intertwining and word order in Ancient Greek". In: *Journal of Greek Linguistics* 12, pp. 5–28.

Andrews, A. D. (1971). "Case Agreement of Predicate Modifiers in Ancient Greek". In: *Linguistic Inquiry* 2.2, pp. 127–151.

Barrett, M. (2025). *ggdag: Analyze and Create Elegant Directed Acyclic Graphs*. R package version 0.2.13.9000. URL: https://github.com/r-causal/ggdag.

Boas, E. v. E. et al. (2019). *The Cambridge Grammar of Classical Greek*. Cambridge: Cambridge University Press.

Comrie, B. (1975). "Polite Plurals and predicate agreement". In: *Language* 51, pp. 406–418.

Corbett, G. G. (1979). "Agreement Hierarchy". In: *Journal of Linguistics* 15.2, pp. 203–224.

— (2006). *Agreement*. Cambridge: Cambridge University Press.
Dik, H. (1995). *Word Order in Ancient Greek: a pragmatic account of the word order variation in Herodotus*. Amsterdam: J. C. Gieben, Publisher.
— (2007). *Word Order in Greek Tragic Dialogue*. Oxford: Oxford University Press.
Kühner, R. and B. Gerth (1898). *Ausführliche Grammatik der griechischen Sprache*. Hannover: Hahn.
Lakoff, G. (1970). "Global rules". In: *Language* 46.3, pp. 627–639.
Lancelot, C. (1658). *Nouvelle Methode pour apprendre facilement la Langue Greque: contenant les regles de Déclinaisons, des Conjugaisons, de l' Investigation du Theme, de la Syntaxe, de la Quantité, des Accens, des Dialectes, & des Licences Poëtiques*. Paris: Pierre le Petit.
— (1709). *Nouvelle Methode pour apprendre facilement la Langue Latine: contenant les regles de Genres, des Déclinaisons, des Préterits, de la Syntaxe, de la Quantité, & des Accens*. Paris: Denys Marietti.
Levin, M. (2001). *Agreement with collective nouns in English*. Lund studies in English, 103. Lund: Lund University.
Matić, D. (2003). "Topic, focus, and discourse structure: Ancient Greek Word Order". In: *Studies in Language. International Journal sponsored by the Foundation "Foundations of Language"* 27.3, pp. 573–633. doi: https://doi.org/10.1075/sl.27.3.05mat.
McElreath, R. (2020). *Statistical Rethinking*. Boca Raton: Taylor & Francis Group, LLC.
Pearl, J. (2009). *Causality: Models, Reasoning, and Inference: Models, Reasoning, and Interference*. Cambridge: Cambridge University Press.
Quicoli, A. C. (1982). *The Structure of Complementation*. Ghent: E. Story-Scientia.
R Core Team (2013). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria. url: http://www.R-project.org/.
Sevdali, C. (2007). " Infinitival clauses in Ancient Greek: overt and null subjects, the role of case and focus ". PhD thesis. Cambridge: University of Cambridge.
— (2013). "Case Transmission beyond Control and the Role of Person". In: *Journal of Historical Syntax* 2.4, pp. 1–52.
Spyropoulos, V. (2005). "The syntax of Classical Greek infinitive". In: *Universal Grammar in the Reconstruction of Ancient Languages*. Ed. by K. Kiss. Berlin: Mouton de Gruyter, pp. 295–337.
Stan Development Team (2025a). *Stan Reference Manual*. Version v2.37.0-rc2. url: https://mc-stan.org.
— (2025b). *Stan User's Guide*. Version v2.37.0-rc2. url: https://mc-stan.org.
Tantalou, N. (2003). "Infinitives with overt subjects in Classical Greek". In: *Μελέτες για την ελληνική γλώσσα: πρακτικά της 23ης ετήσιας συνάντησης του*

τμήματος φιλολογίας του Αριστοτελείου Πανεπιστημίου Θεσσαλονίκης, 17-19 Μαΐου, 2002. Θεσσαλονίκη: Αφοί Κυριακίδη, pp. 358–365.

TEXTOR, J. et al. (2016). "Robust causal inference using directed acyclic graphs: the R package 'dagitty'". In: *International Journal of Epidemiology* 6.45, pp. 1887–1894.

VATRI, A. (2020-2022). *Diorisis Search 3.11*. Version 3.11. URL: https://www.crs.rm.it/diorisissearch/ (visited on 02/28/2023).

VATRI, A. and B. MCGILLIVRAY (2018). "The Diorisis Ancient Greek Corpus". In: *Research Data Journal for the Humanities and Social Sciences* 3.1, pp. 55–65. URL: https://brill.com/view/journals/rdj/3/1/article-p55_55.xml.

— (2020). " Lemmatization for Ancient Greek: An experimental assessment of the state of the art ". In: *Journal of Greek Linguistics* 20.2, pp. 179–196. DOI: https://doi.org/10.1163/15699846-02002001.

WICKHAM, H. et al. (2019). "Welcome to the tidyverse". In: *Journal of Open Source Software* 4.43, p. 1686. DOI: 10.21105/joss.01686.